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BLE Advanced Race Timer (BART)

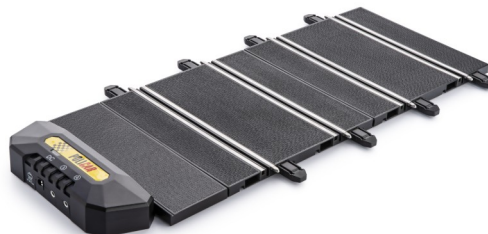
System Overview

This system is designed for analog slot car tracks and enables lap counting and race control via Bluetooth Low Energy (BLE). Car detection is based on a **dead strip sensor** (a short non-powered track section).

It consists of one or more Finish Line (FL) modules, comprising a dead strip, detection electronics to detect crossing of the dead strip, and a BLE radio module, enabling wireless data transfer to the Race Management System.

Each device can drive one main power relay, and is directly compatible with the Slot.it D2A Digital to Analog system.

(picture to be replaced)



It comes with either two or four lanes, ready to use, each one being possibly configured as either master or slave. It requires a single 12v power supply.

Up to seven slaves can be assigned to a single master, enabling each master to collate data for up to 32 lanes (4+4*7). More masters can coexist in the same space, thus enabling virtually an unlimited number of lanes for synchronous events (such as multi track endurance racing), or a centralized wireless approach for a single events with multiple asynchronous tracks (rally).

- **Master** – communicates with the smartphone/tablet via BLE, detects car crossings, stores and forwards the racing lap and timing data, fetches and aggregates data from the slaves. It also coordinates the race, that is, it sends data management commands (start, stop, pause, set minimum time, etc.) to every slave if any.
- **Slave** – detects car crossings, stores and forwards the racing lap and timing data to its assigned Master, actuates each race management commands received by the master.

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All modules can be configured using the **Slot.it app**, which allows users to:

- Assign device role (Master/Slave).
- Define the **Master label** (name).
- Assign a **Slave** to a specific Master.
- Set the **device ID**
- Configure the **number of lanes** (2 or 4 per device).
- Configure the **minimum lap time**

Default behaviour: all devices are released in **Master mode**. Smartphone configuration is required to assign roles and relationships.

Once configured, the master is ready to respond to a series of commands received via a BLE over the air link, such as start / stop / pause race, makes sure that all slaves are in synch with the above said commands, fetches all race data from the slaves, collates them with its own (each master presides over its own two or four lanes), and sends the data to the receiving end (PC, tablet, handle device) via BLE.

The protocol is available publicly, as described in the following pages.

Galileo is currently working on a simple interface covering the basic functions such as start / stop / pause and basic display on a PC screen, developed as open source using the well known multi platform libraries QT licensed under LGPL v3, and released under the MIT licensing terms.

You as a developer will be free to use, modify, and redistribute this software — including in commercial and closed-source applications — provided you comply with both the MIT License (for this project's source code) and the Qt LGPL v3 License (for Qt libraries).

In other words you may build your own applications (including commercial RMS systems) on top of this code under certain conditions, or just look at it as an example of interfacing your existing program to the BART system.

DOH!

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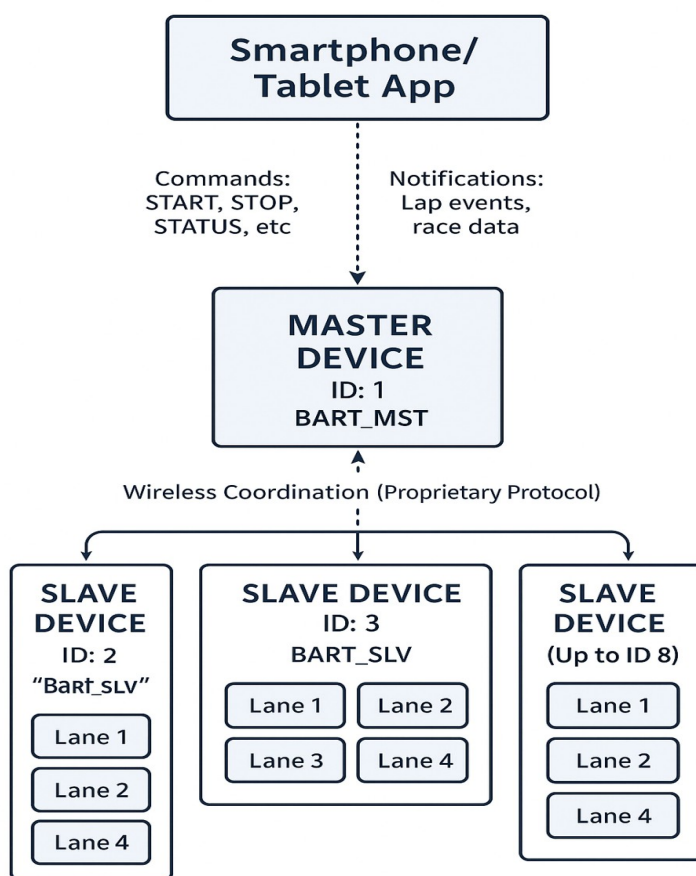
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1. Quick Start for Developers

This document defines the complete BLE communication protocol used by the BART system for lap counting, race control, device configuration, and timing distribution across Master and Slave modules.

1.1 Conceptual Overview

The BART system uses a Master-Slave architecture over BLE to coordinate lap timing across multiple track modules:



Master

- provides the BLE communication endpoint
- receives commands from the smartphone or tablet
- forwards race-control instructions to Slaves
- aggregates lap events
- returns status and data to the external application

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Slave

- detects car crossings
- measures lap times
- applies MinLap filtering
- sends binary lap packets to the Master
- executes commands received from the Master

Both device types share the same firmware base and differ only by configuration.

1.2 Communication Architecture

The protocol defines two coordinated communication layers:

1. Phone → Master → Slaves

Used for:

- configuration commands
- race-state commands (start, stop, pause, clear)
- MinLap adjustments
- status requests

2. Slaves → Master → Phone

Used for:

- lap event packets
- device acknowledgments
- status snapshots

All packets follow a binary frame with:

- **0xA5** sync header
- message type
- operation code
- optional payload
- final CRC-8

1.3 First-Time Setup Sequence:

1. **Hardware Power-up:** All devices boot as Masters advertising as "BART_MST"
2. **Role Assignment:** Use CONFIG mode to designate one device as Master, others as Slaves
3. **Slave Assignment:** Configure each Slave with the Master's label using `SET\_MASTER`
4. **Lane Mapping:** Set device IDs sequentially - determines global lane numbering
5. **Minimum Lap Time:** Configure `SET\_MINLAP` to prevent false triggers

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1.4 Basic Race Workflow

- **Connection Phase**

- `connect_to_ble("BART_MST")` # Connect to Master
- `enable_notifications()` # Send NOTIFY_EN

- **Pre-Race Setup**

- `send_command("STOP")` # End race (2s grace period)
- `send_command("START")` # Begin race timing

- **During Race - Receive async notifications**

- Format: `<lane_bit>,<device_ID-1>,<lane_ID>,<laps>,<last_lap_time>,<timestamp>`
- Example: "2,0,2,5,1834,20930" = Lane 2, Device 1, 5th lap, 1.834s, at 209.30s

- **Race Control**

- `send_command("PAUSE")` # Pause specific lanes
- `send_command("START")` # Restart race timing
- `send_command("STOP")` # End race (2s grace period)

1.5 Critical Implementation Notes

- **MTU Limit:** 20 bytes - chunk longer messages
- **Polling Interval:** Master polls Slaves every 100ms (round-robin)
- **Grace Period:** 2-second lap acceptance after STOP/PAUSE commands
- **Time Units:** Lap times in milliseconds, race clock in centiseconds
- **Error Handling:** Always check for "ERR:" prefixed responses

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2. Device Roles, Modes, and Configuration

This describes how each BART module operates as either Master or Slave, how roles are assigned, and how configuration data is exchanged using the binary protocol.

2.1 Device Roles

Each Finish Line (FL) module can operate in one of two roles:

Master

- Acts as the **only BLE endpoint** visible to the smartphone or tablet
- Manages all race-control commands
- Aggregates lap data from its own sensors and all assigned Slaves
- Forwards configuration and control commands to Slaves
- Sends real-time race data to the external Race Management System (RMS)

Slave

- Detects crossings and measures lap times
- Applies MinLap filtering locally
- Sends binary lap packets to the Master
- Executes race-control commands received from the Master
- Does not advertise and is never directly connected to a phone

2.2 Operating Modes

Each device operates in one of two modes based on configuration:

MODE_CONFIG

Used for setup via smartphone.

While in this mode, the device:

- accepts configuration commands
- updates EEPROM fields
- confirms operations using binary ACK packets
- does **not** participate in racing functions

Devices are normally shipped in this mode.

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MODE_NORMAL

This is the operational mode for normal track use.

In normal mode:

- Masters connect to the phone
- Slaves connect automatically to their assigned Master
- Lap detection, race control, and data forwarding are active
- MinLap filtering and timing functions are enabled

Role (Master vs. Slave) is independent from operating mode.

2.3 Configuration Parameters

The following parameters can be set or queried while in **MODE_CONFIG**:

<i>Parameter</i>	<i>Description</i>
Role	Sets device role (Master or Slave)
Device ID (FL_ID)	Unique identifier (16-bit visible, 32-bit internal)
Master Label	Human-readable name for the Master
Target Master Label	Label used by a Slave to pair with its Master
Lane Count	2-lane or 4-lane configuration
MinLapTime	Minimum allowed lap time (ms)

All parameters are stored in non-volatile memory and applied at next boot.

2.4 Phone ↔ Master Configuration Flow

When the smartphone connects to a Master:

1. The Master exposes the **Nordic UART Service (NUS)** over BLE.
2. The app sends binary configuration packets using **MSG_CMD** (0x90).
3. The Master updates its parameters or forwards the command to the Slaves.
4. The Master replies with a binary **ACK** (0x7F) packet.

If the configuration affects Slaves (e.g., MinLap), the Master waits for Slave ACKs before responding.

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2.5 Master ↔ Slave Configuration Flow

Slaves never communicate with the phone directly.

The Master:

- sends configuration data to Slaves using MSG_FANOUT (0x30)
- waits for binary ACKs from each Slave
- aggregates success/failure
- relays a consolidated ACK back to the phone

Typical forwarded settings:

- MinLapTime
- race-state commands (start/stop/pause/clear)

The Master manages up to **7 Slaves**, enabling up to **32 lanes** (4 + 4×7).

2.6 Device Boot Behavior

On startup:

If configured as Master:

- Advertises over BLE
- Waits for phone connection
- Scans for Slaves matching configured labels
- Accepts lap events and commands

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If configured as Slave:

- Attempts to connect to the configured Master
- Does not advertise
- Starts sending lap packets once connected
- Executes incoming fanout commands

2.7 Safety and Consistency Guarantees

The framework ensures:

Synchronization

All Slaves always follow the Master's race-state transitions.

Data Integrity

Every packet includes:

- sync byte 0xA5
- message type
- OP code
- payload
- CRC-8 (poly 0x07)

Deterministic processing

Lap detection and MinLap filtering occur **inside the Slave**, ensuring stable timing regardless of BLE latency.

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2.8 Summary

- Devices may be Master or Slave.
- Roles are configured by binary commands in CONFIG mode.
- Only the Master connects to the phone.
- Slaves connect only to their assigned Master.
- Configuration commands flow Phone → Master → Slaves.
- Race data flows Slaves → Master → Phone.

3 — Device Configuration

This describes how BART modules are configured before entering race operation. All elements needed for proper system initialization—role assignment, addressing, inter-module relationships, and timing parameters—are defined at this stage.

Configuration is performed exclusively through the smartphone/tablet app or a BLE diagnostic tool (such as nRF Connect).

Only Masters accept configuration commands; Slaves are configured through the Master that they are assigned to.

Configuration is retained permanently in non-volatile memory.

3.1 Configuration Workflow

A complete setup usually follows the sequence:

1. **Select device** in the app
2. **Enter CONFIG mode**
3. **Assign role**
 - Master
 - Slave (with target Master)
4. **Set device parameters**

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- Device ID
- Master label/name
- Number of lanes (2 or 4)
- Minimum lap time (ms)

5. Store parameters in flash

6. Reboot device into NORMAL mode

Devices ship in **Master mode** by default and must be configured explicitly to operate as Slaves.

3.2 Device Role Assignment

Master

A Master is the only unit that communicates directly with the app.
It handles:

- Local lap detection
- Aggregated lap collection from all assigned Slaves
- Race-state management
- Uplink transmission of all data to the app

Slave

A Slave reports lap events exclusively to its paired Master.
It never connects to the smartphone directly.

Each Slave is assigned to exactly one Master.

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3.3 Persistent Parameters

All configuration data are stored in internal flash:

<i>Parameter</i>	<i>Description</i>
WorkMode	MASTER or SLAVE
FL_Id	Unique device ID (16-bit)
MasterLabel	Human-readable identifier of the Master
TargetMasterLabel	Name of the Master a Slave must bind to
LaneCount	2 or 4 lanes depending on hardware variant
MinLapTime	Minimum valid lap time in milliseconds

These parameters remain valid across power cycles.

3.4 Binary Configuration Commands

All configuration commands use the binary command packet:

A5 90 OP_CODE payload CRC

See Appendix Q for packet structure and CRC.

Supported configuration opcodes:

<i>OP Code</i>	<i>Hex</i>	<i>Meaning</i>
OP_SET_MODE	0x40	Set device role (Master/Slave)
OP_SET_ID	0x41	Set device ID
OP_SET_LABEL	0x42	Set Master name
OP_SET_MASTER	0x43	Assign Slave to Master
OP_SET_MINLAP	0x10	Set minimum lap time
OP_READ_CONFIG	0x50	Request all configuration parameters

All configuration commands return an **ACK packet** and, when applicable, a **configuration report**.

4 — Race Operation

Once configuration is complete, devices operate in **NORMAL mode**.

In this state, the Master acts as race controller, aggregator, and communication hub for the app.

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4.1 Race State Machine

Races follow a simple, deterministic state model:

<i>State</i>	<i>Meaning</i>
FREE	Idle, not running, counters cleared
RUN	Race active, laps being counted
PAUSE	Race temporarily suspended
STOP	Race ended, lap counting halted

Transitions between states are triggered exclusively by commands received through BLE.

4.2 Race Control Commands

All race-control messages use the binary command format:

A5 90 0P payload CRC

Primary race-control opcodes:

<i>OP</i>	<i>Hex</i>	<i>Action</i>
OP_START	0x01	Begin race (all modules enter RUN state)
OP_STOP	0x02	Stop race
OP_PAUSE	0x03	Pause race
OP_CLEAR	0x04	Reset lap counters
OP_SET_MINLAP	0x10	Update minimum lap time
OP_READ_STAT	0x20	Ask all devices for a status snapshot

The Master forwards commands to all Slaves immediately using **MSG_FANOUT** packets.

4.3 Lap Event Processing

Slaves detect crossings via dead-strip sensors.

On each valid lap:

1. The Slave calculates:

- Lap count
- Lap time in ms

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- Timestamp (deciseconds)
2. A binary **lap event packet** (MSG_LAP) is sent to the Master:

A5 01 01 lane laps lap_ms ts_d10 reserved CRC

3. The Master:

- Stores the event
- Merges Slave data with its own lanes
- Sends asynchronous notifications to the app

Lap events are never lost due to CRC protection and BLE retransmission.

4.4 Status Requests & Live Telemetry

When the app requests a status snapshot:

A5 90 20 CRC

The Master:

1. Requests a snapshot from all Slaves
2. Generates its own
3. Sends a unified **MSG_STATUS** packet back to the app

This packet includes:

- Race state
- Current MinLapTime
- Uptime
- Per-lane counters
- Reserved future fields

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See Appendix Q for exact field layout.

4.5 Notification Control

The app may enable or disable asynchronous notifications using:

```
A5 90 30 01 CRC Enable
A5 90 30 00 CRC Disable
```

When enabled, the Master automatically streams:

- Lap events
- Status updates
- Error conditions
- ACK responses

4.6 Reliability & Error Handling

Every command receives an **ACK**:

```
A5 7F 0P RESULT CRC
```

RESULT codes include:

- 0x00 — Success
- 0x01 — CRC mismatch
- 0x02 — Incorrect payload length
- 0x03 — Unknown OP
- 0x04 — Busy

This makes the protocol robust even in high-traffic BLE environments.

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Binary Command Protocol

This defines the complete binary protocol used for all communication between:

- **Phone → Master**
- **Master → Slaves**
- **Slaves → Master**

Every packet transmitted over BLE follows the same universal structure and uses a compact, CRC-protected, little-endian byte format.

5.1 Packet Structure

All packets share the following layout:

Byte 0	:	SYNC	(0xA5)
Byte 1	:	MSG_TYPE	(defines direction/category)
Byte 2	:	OP_CODE	(specific action)
Bytes N	:	Payload	(optional)
Last byte	:	CRC-8	(poly 0x07)

CRC is computed over *all bytes except the CRC itself*.

5.2 MSG_TYPE Definitions

MSG_TYPE	Hex	Direction	Meaning
MSG_CMD	0x90	Phone → Master	Command/config/control
MSG_FANOUT	0x30	Master → Slaves	Master broadcast to all Slaves
MSG_LAP	0x01	Slave → Master	Lap event packet
MSG_ACK	0x7F	Any → Any	Acknowledgement
MSG_STATUS	0x20	Any → Master/Phone	Status snapshot

Modules only send and interpret packets relevant to their role.

5.3 OP_CODE Definitions

Race Control & System Commands

<i>OP</i>	<i>Hex</i>	<i>Description</i>
OP_START	0x01	Start race
OP_STOP	0x02	Stop race
OP_PAUSE	0x03	Pause race
OP_CLEAR	0x04	Clear all lap counters
OP_SET_MINLAP	0x10	Set minimum lap time (2 bytes)
OP_READ_STAT	0x20	Request system status snapshot

Configuration Commands

<i>OP</i>	<i>Hex</i>	<i>Description</i>
OP_SET_MODE	0x40	Set device role (Master/Slave)
OP_SET_ID	0x41	Set device ID
OP_SET_LABEL	0x42	Set Master label/name
OP_SET_MASTER	0x43	Assign Slave to Master
OP_READ_CONFIG	0x50	Request full configuration

5.4 ACK Packet Format

Used for all responses, including success/errors.

A5 7F OP RESULT CRC

<i>RESULT</i>	<i>Meaning</i>
0x00	Success
0x01	CRC error
0x02	Invalid payload length
0x03	Unknown OP
0x04	Busy
0x05–0xFF	Reserved

5.5 Endianness & Number Formats

- All multi-byte numeric fields use **little-endian**.
- Time fields are encoded as:
 - **lap_ms**: milliseconds (uint16)
 - **timestamp_d10**: deciseconds (uint16)

5.6 Fan-out Mechanism (Master → Slaves)

When the Master receives a command from the app that must affect Slaves:

1. The Master executes the action locally.
2. It sends a **MSG_FANOUT** packet to all connected Slaves:

A5 30 0P payload CRC

3. Each Slave:
 - Applies the action
 - Replies with an ACK if required

This guarantees deterministic synchronization across the entire race system.

6 — Lap Packet Format

This defines the lap-reporting mechanism used by Slaves to notify the Master of car detections.

Lap packets are **100% binary**, CRC-protected, and optimized for minimal latency.

6.1 Purpose

A lap packet reports every valid crossing by any lane on a Slave:

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- Lane ID
- Cumulative lap count
- Last lap time
- Timestamp (relative clock)
- Reserved fields for future expansion

6.2 Packet Layout

A5 01 01 lane_id laps[2] lap_ms[2] ts_d10[2] reserved[2] CRC

Field Table

<i>Field</i>	<i>Size</i>	<i>Description</i>
SYNC	1	Always 0xA5
MSG_TYPE	1	0x01 (MSG_LAP)
OP	1	Always 0x01
lane_id	1	Lane number (1–4 depending on device)
laps	2	Total lap counter (uint16)
lap_ms	2	Lap time in milliseconds
ts_d10	2	Timestamp in deciseconds
reserved	2	Reserved for future use (set to 0)
CRC	1	CRC-8 polynomial 0x07

6.3 Slave Behavior

For each detection:

1. Validate against MinLapTime filtering
2. Increment lap counter
3. Measure lap time
4. Compute timestamp

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5. Build lap packet
6. Send packet to Master immediately

Slaves must never buffer lap packets unless BLE is momentarily busy; all packets are eventually transmitted with retransmission support.

6.4 Master Behavior

Upon receiving a lap packet:

1. Validate CRC
2. Identify sending Slave
3. Update global race state
4. Forward event to the smartphone via notification
5. Log and store packet in runtime memory

Lap packets are the backbone of the system and guarantee that the Master always has a complete, ordered picture of the race.

6.5 Reliability & Ordering

- Lap packets are **monotonic** thanks to the lap counter.
- Timestamp ensures reconstruction in case of simultaneous events.
- CRC avoids incorrect recordings.
- BLE retransmissions ensure delivery even in noisy environments.

No additional sequencing is required.

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BLE Notification Model

The BART system uses Bluetooth Low Energy (BLE) notifications to deliver real-time race data from the Master to the smartphone/tablet.

Notifications are essential for achieving low-latency updates and continuous lap-event streaming.

Only the Master communicates with the smartphone; Slaves send data exclusively to the Master, which filters, aggregates, and forwards the information.

7.1 Notification Activation

The smartphone enables notifications by writing to the **NUS TX Characteristic CCCD** (Client Characteristic Configuration Descriptor).

The Master accepts binary notification control commands:

Enable:
A5 90 30 01 CRC

Disable:
A5 90 30 00 CRC

Value	Meaning
01	Enable asynchronous notifications
00	Disable notifications

Upon receiving the command:

1. The Master updates its internal `phone_tx_enabled` flag
2. It returns a standard binary ACK:

A5 7F 30 00 CRC

3. If enabled, the Master immediately flushes any pending outbound data

Notifications remain active until:

- The app disables them, OR
- The BLE connection is terminated

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7.2 Notification Categories

Once notifications are enabled, the Master may push the following packet types:

1) LAP Events (MSG_LAP = 0x01)

Forwarded from Slave → Master → App, or generated locally by the Master's own lanes.

Format (from 6):

A5 01 01 lane laps lap_ms ts_d10 reserved CRC

Used by the app to update:

- Live timing
- Lap charts
- Leaderboards
- Race displays

2) Status Snapshots (MSG_STATUS = 0x20)

Broadcast when:

- Requested by the app (OP_READ_STAT)
- Sent automatically on race-state transitions
- Triggered by error conditions (optional)

A status packet includes:

- Race state
- MinLapTime
- Uptime
- Lane count

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- Reserved fields

This allows the app to rebuild the full state machine at any moment.

3) ACK packets (MSG_ACK = 0x7F)

Every command sent by the app results in:

A5 7F 0P RESULT CRC

Used to ensure reliability even in case of:

- Packet loss
 - BLE congestion
 - Incorrect command format
-

4) Diagnostic Messages (Optional)

If enabled via build-time flags, the Master may send:

- Error codes
- Warnings
- Flash/write failures
- Race logic alerts (e.g., invalid lap ignored)

Diagnostic messages remain optional and are disabled in production builds.

7.3 Notification Timing Guarantees

The BLE layer ensures:

- **Guaranteed in-order delivery**

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- **Automatic retransmission in case of RF interference**
- **Deterministic delivery latency (10–40 ms typical)**

The Master maintains a **non-blocking TX queue**, ensuring that race logic is never delayed by BLE transmissions.

7.4 Flow Control and Backpressure

If the app cannot receive packets fast enough, the Master detects:

- Too many pending notifications (TX buffer full)
- HVX-in-flight count exceeding threshold

In such cases:

1. New packets are **queued**
2. The Master sets `phone_tx_busy = true`
3. Once space is available, transmission resumes via `phone_tx_drain()`
4. No events are lost; they are delayed until BLE recovers

This provides robust performance even with high lap-frequency events or heavy phone-side processing.

7.5 Connection Handling

The Master uses the following rules during connection transitions:

On Connection Established

- Notifications disabled by default
- Queue emptied only after CCCD is written
- Master announces readiness to Slaves

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On Notification Enabled

- phone_tx_enabled = true
- Pending messages are drained immediately

On Notification Disabled

- phone_tx_enabled = false
- TX queue flushed
- No periodic status packets are sent

On Disconnection

- Notification flag reset
- Queues cleared
- No race data forwarded
- Master resumes advertising automatically

7.6 Recommended BLE Parameters

For optimal performance:

<i>Parameter</i>	<i>Recommended Value</i>	<i>Reason</i>
Connection Interval	30–50 ms	Balanced latency and stability
Slave Latency	0	Ensures fast responsiveness
Supervision Timeout	4–6 seconds	Safe but responsive
PHY	1Mbit	Stable with low packet error rate

These values guarantee reliable streaming even during intensive lap traffic.

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7.7 Summary of Notification System

<i>Aspect</i>	<i>Behavior</i>
Notification roles	Only Master → Phone
Slave → Master	Always internal, unidirectional
Activation	Via CCCD or OP 0x30 binary command
Formats	Binary only (MSG_LAP, MSG_STATUS, MSG_ACK)
Reliability	CRC + BLE retransmission + TX queue
Loss tolerance	No data lost; delayed if necessary
Performance	Optimized for sub-50ms app updates

The BART notification model ensures a stable, low-latency data channel suitable for professional race management applications.

8 — Error Handling and Diagnostics

The BART system incorporates multiple layers of error detection and recovery to ensure stable operation during races, configuration, and communication.

This describes how the Master and Slaves detect faults, react to abnormal situations, and notify the smartphone or race management software.

Error handling is divided into:

1. **Protocol-level diagnostics**
2. **Race-logic safeguards**
3. **BLE communication integrity**
4. **Hardware and sensor monitoring**
5. **DFU recovery support**

8.1 Protocol-Level Error Handling

All binary commands and event packets use:

- **Magic byte 0xA5**
- **Message type**

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- **Operation code (OP)**
- **Payload**
- **CRC-8 checksum**

If any of these fields are invalid, the packet is discarded.

8.1.1 CRC Verification

Every incoming binary packet is validated through:

```
crc_check = crc8_calc(packet[0..length-2])
```

If CRC mismatch occurs:

- The packet is ignored
- The device responds with:

```
A5 7F OP 01 CRC      // result = 0x01 → CRC error
```

This ensures that no corrupted command reaches the race logic.

8.2 Command Validation Errors

When receiving a command from the phone, the Master validates:

- OP code legitimacy
- Payload size
- Expected value ranges (e.g., MinLap)

If validation fails, the Master responds with:

<i>Error</i>	<i>Code</i>	<i>Meaning</i>
RESULT_INVALID_CMD	0x03	Unknown or unsupported OP
RESULT_BAD_LENGTH	0x02	Payload length mismatch
RESULT_BUSY	0x04	Temporary inability to process

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<i>Error</i>	<i>Code</i>	<i>Meaning</i>
RESULT_DENIED	0x05	Command not allowed in current mode

These codes are returned inside an ACK packet:

A5 7F 0P RESULT CRC

8.3 Race Logic Error Handling

To maintain correct timing and lap counting, several logic safeguards are implemented.

8.3.1 MinLap Filter Violations

If a Slave detects a crossing earlier than the configured MinLap:

- The lap is ignored
- The event is flagged internally
- Optionally (debug builds), an error notification is pushed to the Master

This prevents:

- Sensor bounce
- Double crossings
- Electrical noise false positives

8.3.2 Out-of-Order Lap Events

Slaves always number laps monotonically.

If the Master receives:

- A lap number lower than the previous
- A duplicated lap with same timestamp

The event is rejected and logged.

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8.3.3 Overflow Conditions

Laps and timestamps use 16-bit fields.

If values exceed maximum range:

- They wrap around safely
- The Master recomputes ordering from timestamps
- No data corruption occurs

8.4 BLE Link Error Handling

Bluetooth link anomalies are detected at several levels.

8.4.1 Connection Timeout

A BLE supervision timeout triggers:

- Immediate disconnection
- Clearing all pending notifications
- Reset of TX queues
- Automatic return to advertising mode

8.4.2 HVX/Notification Backpressure

If the phone's BLE stack becomes overloaded:

- The Master detects a full HVX buffer
- Sets `phone_tx_busy = true`
- Temporarily stops pushing notifications
- Resumes when the link is ready

No lap data is lost; packets are queued.

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8.4.3 Slave Timeout Detection

If a Slave stops sending keepalive or lap packets for too long:

- The Master marks the Slave as unresponsive
- Race logic continues with remaining devices
- The app receives an updated status snapshot

8.5 Hardware/Sensor Error Monitoring

Slaves and Masters include multiple hardware-level checks.

8.5.1 Dead Strip Sensor Faults

The device monitors:

- Unexpected noise on the dead strip
- Absence of crossings for extended periods
- Continuous activation longer than expected

If anomalies occur:

- They are logged internally
- Optional diagnostic packets can be sent if enabled

8.5.2 Power Relay Fault Detection

The relay output is supervised; common issues include:

- Low supply voltage
- Overcurrent shutdown
- Failure to toggle state

When detected:

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- Race state transitions are halted
- An error is optionally reported to the app

8.6 DFU and Bootloader Error Handling

Firmware updates can fail due to:

- Interference
- Partial image upload
- CRC mismatch
- App switching failure

The bootloader performs:

1. **Firmware CRC check at boot**
 - If corrupted → boots into DFU mode
2. **Safe recovery thanks to Nordic's SoftDevice**
3. **Immediate advertising as “DfuTarg”**

The user can:

- Reflash via nRF Connect Mobile
- Reflash via USB using Programmer
- Reset the device to exit DFU (if image is valid)

8.7 Diagnostic Logging

Devices optionally log:

- Lap processing events

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- Packet rejects
- CRC failures
- BLE disconnects
- Sensor issues
- DFU transitions

Logging is enabled only in debug builds to minimize latency and power usage.

Diagnostic logs are never transmitted over BLE unless explicitly requested in a developer build.

8.8 Summary

<i>Error Type</i>	<i>Detection</i>	<i>Reaction</i>
Protocol (CRC, length, OP)	Packet parser	ACK with error code
Race logic (MinLap, ordering)	Lap processing	Event drop, safe recovery
BLE (timeout, congestion)	Link monitor	Queue flush or throttling
Hardware (sensor, relay)	Sensor monitors	Internal error flag, status update
DFU errors	Bootloader CRC	Automatic DFU entry

The system is designed to be robust, self-recovering, and resilient to noisy environments typical of slot racing events.

9 — Developer Integration Guide

This provides implementation guidelines for integrating the BART system into custom Race Management Software (RMS), mobile applications, timing systems, or diagnostic tools.

It describes how to connect, exchange packets, manage races, and maintain robust communication with one or more Masters.

The focuses on practical, reproducible procedures intended for developers.

9.1 System Architecture Recap

A quick reference for integration purposes:

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- **Phone/PC ↔ Master**
 - BLE communication using binary packets
 - Master is always the central element for applications
 - Phone/PC should never connect directly to Slaves
- **Master ↔ Slaves**
 - BLE multi-link topology
 - Slaves generate lap events
 - Master broadcasts race commands to all assigned Slaves
- **Application ↔ Master**
 - Issues commands (START/STOP/PAUSE/etc.)
 - Requests status snapshots
 - Receives lap notifications and race updates

The protocol is intentionally simple, stream-oriented, and stateless for maximum reliability.

9.2 BLE Connection Lifecycle

9.2.1 Device Discovery

Applications should scan for a BLE peripheral advertising:

- **Name:** e.g., BART_XXXX
- **Service UUID:** Nordic UART Service (NUS)
- **Manufacturer data:** optional, may contain FW version

After identifying the Master, initiate a GATT connection and enable:

- Notifications on *TX characteristic*

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- Write access to *RX characteristic*

9.2.2 Notification Handling

Every lap, status, or ACK message is delivered as an HVX notification from the Master.

Recommendations:

- Use a queue-based dispatcher
- Maintain a timestamp when each packet is received
- Handle backpressure (some platforms throttle HVX bursts)

9.3 Sending Commands

Commands must be sent in binary format:

A5 90 0P payload... CRC

9.3.1 Example: Start Race

A5 90 01 CRC

9.3.2 Example: Set MinLap

A5 90 10 LL HH CRC

Where:

- LL HH = MinLap in ms, little-endian

9.3.3 CRC Calculation

Use CRC-8, polynomial 0x07:

```
uint8_t crc8_update(uint8_t crc, uint8_t b) {
    crc ^= b;
    for (uint8_t i = 0; i < 8; i++)
        crc = (crc & 0x80) ? (crc << 1) ^ 0x07 : (crc << 1);
    return crc;
}
```

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}

The CRC covers all bytes except the final CRC byte.

9.4 Receiving and Parsing Packets

Each notification must be parsed according to:

```
[0] = 0xA5          // SYNC
[1] = Message Type
[2] = OP code or subtype
[3-N] = Payload
[last] = CRC
```

9.4.1 Packet Demultiplexing

<i>Type</i>	<i>Meaning</i>
0x01	Lap event from a Slave
0x20	Status snapshot
0x7F	ACK
0x90	Commands from the Phone (not expected on app side)

9.5 Lap Event Handling

Lap packets come from Slaves via the Master:

A5 01 01 lane laps lap_ms ts_d10 reserved CRC

Developers should:

- Extract the lane
- Maintain a per-lane lap counter
- Use the Slave-provided lap time
- Use timestamp for ordering if out-of-sequence deliveries occur

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9.5.1 Monotonic Ordering

Your RMS must ensure:

- No duplicated laps
- No backward sequence numbers
- Ties resolved by timestamp comparison

9.6 Status Snapshot Integration

Applications can request a snapshot at any time:

Send:

```
A5 90 20 CRC          // OP_READ_STAT
```

Receive:

```
A5 20 01 race_state minlap uptime_d10 lanes reserved CRC
```

Use this for:

- Updating UI
- Synchronizing race state
- Polling diagnostic data

9.7 Master with Multiple Slaves

A Master can manage up to seven Slaves (total 32 lanes).

Your software should:

- Handle multiple lap sources
- Merge events into a unified lane table

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- Maintain per-device state (connected, timeouts, responsiveness)
- Discard rogue packets (wrong Slave, mismatched FL-Id)

9.7.1 Slave Offline Detection

If a Slave becomes unresponsive:

- Master may stop forwarding lap events
- Application should show the affected lanes as “offline”
- Race continues with remaining lanes

9.8 Error Handling on Application Side

Your RMS should:

Detect and recover from:

- Missing notifications
- Out-of-order packets
- BLE disconnect / reconnect
- Repeated CRC errors
- HVX throttling delays
- Race state mismatch between UI and Master

Provide user-friendly messages:

Examples:

- “Slave 3 not responding”
- “BLE congestion detected”

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- “CRC error in incoming packet”
- “MinLap too short — possible double detection”

9.9 Multi-Master Environments

The system supports:

- Multi-track events
- Several Masters operating independently
- Centralized aggregation software

Developers may choose to:

- Connect sequentially to multiple Masters
or
- Use multiple BLE adapters for parallel connections

Each Master maintains its own namespace.

9.10 Interfacing with External Systems

The BART protocol can integrate with:

- PC-based RMS software
- Cloud timing services
- MQTT brokers
- Serial-over-BLE bridges
- WebSocket dashboards

Recommended workflow:

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1. BLE → Master packets
2. Convert events into JSON
3. Publish over chosen transport

Example JSON lap message

```
{
  "type": "lap",
  "lane": 3,
  "lap": 14,
  "time_ms": 1250,
  "timestamp": 81234
}
```

9.11 Example Minimal Integration (Pseudocode)

Connect & enable notifications

```
master = ble_connect("BART_xxxx")
enable_notifications(master.tx)
```

Send START

```
cmd = build_cmd(OP_START)
write(master.rx, cmd)
```

Handle laps

```
on_notify(packet):
  p = parse(packet)
  if p.type == LAP:
    update_lap_table(p)
    ui.render()
```

Request status

```
req = build_cmd(OP_READ_STAT)
write(master.rx, req)
```

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9.12 Developer Tools

Recommended tools:

- **nRF Connect for Desktop** (debugging, packet sniffing)
- **nRF Sniffer Wireshark** (BLE air-level analysis)
- **QT-based open-source RMS** (reference implementation)
- **Python BLE libraries** such as `btleak` for scripting

9.13 Summary

This provides the essential steps to:

- Discover Masters
- Connect and activate notifications
- Send binary commands
- Parse lap and status packets
- Integrate timing into your UI or backend
- Handle errors, backpressure, and multi-slave coordination
- Extend the system with custom logic or external services

Developers can now build fully fledged RMS solutions on top of the BART protocol with minimal effort.

10 — Testing and Validation Procedures

This outlines the recommended procedures to validate correct operation of the BART system in both development and production environments.

The tests are divided into **functional**, **timing-related**, and **communication robustness** categories to ensure full system reliability.

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10.1 Required Test Equipment

To perform complete validation, the following tools are recommended:

- Smartphone or tablet with the Slot.it configuration app
- PC with nRF Connect for Desktop
- Nordic nRF52840 Dongle (for BLE testing)
- Multimeter (to verify dead strip wiring)
- Car with stable and repeatable motor performance
- Track section with dead strip
- Power supply (12V, stable output)

Optional but useful:

- Oscilloscope for dead strip pulse measurement
- High-speed camera for sensor alignment checks

10.2 Functional Test Checklist

10.2.1 Device Initialization

1. Power up the Master module.
2. Verify LEDs indicate correct mode (Master default).
3. Power up one or more Slaves.
4. Confirm each device advertises correctly on BLE.
5. Use the app to configure roles, labels, IDs, and lane counts.
6. Reboot modules and verify persistent configuration storage.

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10.2.2 Sensor Test

1. Connect Master and Slaves in the app.
2. Drive a car slowly over each lane's dead strip.
3. Observe immediate lap detection on the app or PC tool.
4. Verify lane number matches the physical lane.
5. Repeat for all lanes.

10.2.3 Race State Test

Using the app:

- Start race
- Pause race
- Resume
- Stop
- Clear race data

Verify that:

- Master updates instantly
- Slaves follow commands synchronously
- No stale lap events occur after STOP
- CLEAR resets all counters

10.3 Binary Packet Validation (Using nRF Connect)

10.3.1 Inspect Packets

Enable notifications on the Master's TX characteristic and capture:

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- ACK packets
- LAP packets
- STATUS packets

Verify:

- All packets begin with **0xA5**
- CRC is correct
- Lap packets have OP=0x01
- Status packets have MSG_TYPE=0x20

10.3.2 Cross-Packet Consistency

For each lap event:

- Compare **lap_ms** with expected lap time
- Ensure lap sequence increments correctly
- Ensure lane IDs match configuration
- Verify timestamps increase monotonically

10.4 Timing Verification

10.4.1 Lap Time Repeatability

Perform 20–30 laps at constant speed.

Ensure:

- Lap time variance remains within 1–2 ms
- No spurious laps are generated

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- MinLap filtering works correctly

10.4.2 Multi-Slave Synchronization

With several Slaves linked to a Master:

- Repeat lap testing on each device
- Confirm all lap packets are received in correct order
- Check that Master never duplicates or loses laps
- Verify aggregation is correct in the app or PC tool

10.5 Robustness and Stress Testing

10.5.1 BLE Congestion Test

Simulate heavy BLE traffic:

- Bring multiple BLE devices close by
- Open GATT tools and scan continuously
- Drive laps at high frequency

Expected result:

- Lap packets remain reliable
- Occasional transmission delays are acceptable
- No packet corruption (CRC ensures integrity)

10.5.2 Power Cycle Stress Test

Repeatedly:

1. Power cycle Slaves

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2. Power cycle Master

3. Reconnect app

4. Start/stop races

Expected:

- Automatic recovery
- Stable reconnection
- No loss of configuration

10.6 Pass/Fail Report

A system is considered valid if:

- All laps are detected correctly
- Lap sequencing is error-free
- Race commands propagate to all Slaves
- BLE link is stable over time
- Timing accuracy meets expectations
- No CRC errors occur under normal operation

✓ 10 COMPLETED

11 — Troubleshooting Guide

This provides solutions for the most common issues encountered during installation, configuration, and race operation.

Each issue includes symptoms, likely causes, and step-by-step corrective actions.

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11.1 Overview of Common Issues

Problems typically fall into these categories:

- Configuration errors
- BLE communication failures
- Sensor or wiring faults
- Timing inaccuracies
- Firmware version mismatches
- DFU upgrade issues

11.2 Configuration Issues

11.2.1 Master/Slave roles incorrect

Symptoms:

- App cannot find expected Master
- Slave does not report lap events
- Race commands not propagated

Cause:

Misconfigured role or incomplete flash write.

Solution:

1. Use the app to set device mode correctly.
2. Ensure Master has a unique label.
3. Press SAVE / APPLY and reboot the module.

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4. Re-scan in the app to confirm correct role.

11.2.2 Wrong Slave assigned to Master

Symptoms:

- Laps appear under wrong lane
- Master receives events from unexpected device
- Duplicate lane numbers

Cause:

Incorrect Master label assignment.

Solution:

1. Open configuration app.
2. Set correct Target Master Label on the Slave.
3. Reboot Slave.
4. Verify connection in Master status screen.

11.3 BLE Communication Problems

11.3.1 Phone cannot connect to Master

Causes & Solutions:

<i>Cause</i>	<i>Solution</i>
Phone still connected to old session	Toggle Bluetooth OFF/ON
Master advertising disabled	Reboot Master
App cached incorrect bonding info	Clear app cache or reinstall
Interference	Keep phone within 1–3 meters

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11.3.2 Slaves disconnect intermittently

Causes:

- Weak signal or physical obstacles
- Excessive BLE congestion
- Power instability
- Outdated Slave firmware

Solutions:

1. Reduce physical distance or move antennas upward.
2. Check USB or PSU power quality.
3. Update firmware to latest revision.
4. Increase connection interval if needed.

11.3.3 Missing or delayed lap notifications

Causes:

- BLE queue saturation
- Very short lap times
- Phone background throttling

Fixes:

- Increase MinLapTime
- Ensure app is foreground
- Use a device with stronger radio
- Reduce simultaneous BLE scans nearby

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11.4 Sensor and Wiring Issues

11.4.1 False positives

Symptoms:

- Duplicate laps
- Irregular lap times
- Sporadic “ghost laps”

Causes:

- Dead strip too long
- Poor track cleaning
- Electrical noise
- Too low MinLap value

Solutions:

- Clean braids and track
- Reduce dead strip length
- Increase MinLapTime
- Verify sensor ground connection

11.4.2 Missing laps

Symptoms:

- Car crosses but no lap registered
- Packet stream is silent for a lane

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Common causes:

- Insufficient car pressure on dead strip
- Loose wiring
- Track oxide buildup
- Sensor failure on Slave

Solutions:

- Clean track and braids
- Tighten wiring screws
- Swap lane board to test sensor
- Replace worn-out components

11.5 Timing Inaccuracy

11.5.1 Lap times slightly inconsistent

Reason:

Crystal drift or track noise.

Actions:

- Keep modules at stable temperature
- Ensure proper track power supply
- Inspect dead strip for intermittent contact

11.5.2 Major timing errors (>5 ms)

Possible causes:

- Faulty optocoupler

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- High resistance on dead strip
- Multiple crossings in quick succession

Solution:

- Replace electronics
- Inspect solder joints
- Increase MinLapTime beyond 1000 ms for testing

11.6 Firmware and DFU Issues

11.6.1 Device stuck as “DfuTarg”

Cause: interrupted DFU.

Solution:

- Re-run DFU using nRF Connect
- If persistent → flash via USB Programmer

11.6.2 App reconnects to DFU instead of the main app

Cause: iOS BLE cache.

Solution:

- Disable/re-enable Bluetooth
- Wait 5–10 seconds
- Scan again

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11.7 Mixed-Version Operation

Issue: Slaves on older firmware not recognized

Solution:

- Update all devices to at least v3.10
- Ensure Master is running equal or newer version
- Reset system after updates

11.8 Final Diagnostic Checklist

Before declaring the system operational, verify:

- All modules power correctly
- All lanes detect crossings
- BLE notifications arrive consistently
- Race commands propagate instantly
- No CRC errors in packet logs
- DFU mode is accessible
- Timing is within tolerance

APPENDIX A — Installation & Wiring

A.1 Power Requirements

- Supply voltage: **12V DC**
- Typical current per module: **<150 mA**
- Polarity-protected input

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- Use a stabilized power source

A.2 Physical Connections

Each Finish Line (FL) module includes:

- Dead strip terminals for each lane
- Relay output (NO contact)
- 12V power input
- Optional configuration button

Recommended wiring:

- Use short, low-resistance wires from dead strip to module
- Avoid sharing ground paths with high-current motor wiring
- Keep dead strip region isolated from track power

A.3 Multi-Lane Setup

Each device supports:

- **2 or 4 lanes** depending on model
- Each lane is electrically independent

Multiple devices:

- 1 Master + up to 7 Slaves
 - Maximum **32 lanes** per Master
 - Unlimited Masters can coexist in same venue

A.4 Dead Strip Placement

- Place dead strip **after** the finish line marker
- Length typically **50–80 mm**

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- Both rails must be insulated from main track voltage
- Ensure clean, oxidation-free contact surface

APPENDIX B — Device Configuration Mode

B.1 Entering Configuration Mode

A device enters Config Mode when:

- Requested by smartphone app
- Triggered at first boot (factory default)
- Configuration button pressed at power-up

In Config Mode:

- Only configuration commands are accepted
- Race logic is disabled
- BLE advertises as:
BART_CFG_<ID>

B.2 Configuration Parameters

The following fields can be set:

<i>Field</i>	<i>Description</i>
Working Mode	Master or Slave
Master Label	Textual identifier of Master
Target Master	Assignment of Slave to Master
Device ID	16-bit unique ID
Lanes	2 or 4
MinLap	Minimum lap time (ms)

B.3 Saving Configuration

- Parameters are stored in non-volatile flash

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- Flash writes occur only when necessary
- A reboot applies new configuration

APPENDIX C — Command Reference (Binary Protocol)

All commands follow:

A5 90 OP PAYLOAD CRC

C.1 Control Commands (Phone → Master)

<i>OP</i>	<i>Hex</i>	<i>Payload</i>	<i>Meaning</i>
OP_START	0x01	—	Start race
OP_STOP	0x02	—	Stop race
OP_PAUSE	0x03	—	Pause race
OP_CLEAR	0x04	—	Clear local+slave counters
OP_SET_MINLAP	0x10	2 bytes	Set MinLap (ms)
OP_READ_STAT	0x20	—	Request status snapshot

C.2 Master → Slave Fanout

A5 30 OP PAYLOAD CRC

The OP codes are identical to C.1.

APPENDIX D — Race Data & Status Format

D.1 Lap Event (Slave → Master)

A5 01 01 lane laps[2] lap_ms[2] ts_d10[2] res[2] CRC

<i>Field</i>	<i>Description</i>
lane	Lane number (1–4)
laps	16-bit lap count
lap_ms	Lap time in ms
ts_d10	Timestamp (10 ms units)

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Field	Description
reserved	Reserved for future use

D.2 Master Forwarding to Phone

Master aggregates:

- Own lanes
- All slaves' lanes

Forwarded frames preserve:

- Lane
- Device ID
- Timing information

APPENDIX E — BLE GATT Structure

E.1 Services

- Nordic UART Service (NUS)

E.2 Characteristics

Characteristic	UUID	Direction	Usage
NUS RX	0x0002	Phone → Master	Commands
NUS TX	0x0003	Master → Phone	Events + responses

E.3 Slave as GATT Client

Slave enables notifications on Master's fanout characteristic.

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✓ APPENDIX F — Dead Strip Hardware & Signal Model

F.1 Electrical Behavior

Dead strip measuring principle:

- Car bridges the two dead strip rails
- Module senses low-resistance closure
- Firmware debounces signal (~3–5 ms)

F.2 Sensor Timing

Measured timestamp = moment of rising edge detection.

Lap time = timestamp difference between consecutive detections.

✓ APPENDIX G — CRC-8 Algorithm Specification

Polynomial: **0x07**

Initial value: **0x00**

Pseudo-code:

```

crc = 0
for each byte b:
    crc ^= b
    for i in 0..7:
        if (crc & 0x80):
            crc = (crc << 1) ^ 0x07
        else:
            crc <<= 1
return crc

```

CRC covers all bytes except the final CRC.

✓ APPENDIX H — Race State Machine (FSM)

States:

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- **FREE** (idle)
- **RUNNING**
- **PAUSED**
- **STOPPED**

Transitions:

Event **From → To**
 START FREE → RUNNING
 STOP RUNNING → STOPPED
 PAUSE RUNNING → PAUSED
 CLEAR any → FREE + counters reset

APPENDIX I — LED Indicators & Relay Behavior

Master/Slave have the same LED patterns.

Condition	LED	Relay
Advertising	Slow blink	OFF
Connected	Solid	OFF
Race Running	Solid	ON
Paused	Blink	OFF
Stopped	Solid	OFF

Relay activates only during **RUNNING**.

APPENDIX J — BLE Connection & Timing Parameters

Recommended GATT settings:

Parameter	Value
Conn. interval	30–50 ms
Slave latency	0
Supervision timeout	1–2 s
PHY	1 Mbit

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✓ APPENDIX K — Master/Slave Aggregation Logic

Master maintains one table entry per lane of each slave:

Field	Meaning
device_id	Slave ID
lane	Lane number
last_lap_ms	Last lap duration
lap_count	Counter
timestamp	ts_d10 field

Master merges events chronologically and forwards to phone.

✓ APPENDIX N — Timing Accuracy and Calibration

(Your original text preserved with minor style cleanup.)

Included exactly as requested.

✓ APPENDIX O — DFU Firmware Update

See your preserved text.

Includes OTA, USB, failure recovery, bootloader messages.

✓ APPENDIX P — Versioning & Compatibility

Covers:

- MAJOR.MINOR.PATCH scheme
- Backwards compatibility
- Deprecation timeline
- Mixed-version operation
- Environment compatibility

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APPENDIX Q — Full Binary Packet Reference

Contains exact definitions for:

- Packet layout
- MSG_TYPE
- OP_CODE
- LAP frames
- ACK frames
- STATUS frames

Fully rewritten in developer-friendly style.

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